

Robust Electricity Forecasting in Smart Buildings with Missing Data: A Concept Echo State Network Approach

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1. Background

- Accurate electricity forecasting is critical for optimizing energy use in smart buildings and ensuring grid stability.
- Real-world data is messy. Smart buildings frequently have high rates of missing data (15-30%) due to sensor or communication failures.

2. Motivation

- Complex deep learning models such as **LSTM** are widely used for time-series forecasting but often behave as *"black boxes"*, limiting interpretability.
- They typically **struggle to handle missing or incomplete data**, a common issue in smart-building energy datasets.
- These weaknesses lead to **unstable and unreliable forecasts**, especially under conditions of data loss or irregular sampling.
- There is a strong need for a **transparent, lightweight, and resilient** modeling framework that maintains accuracy even with imperfect data.

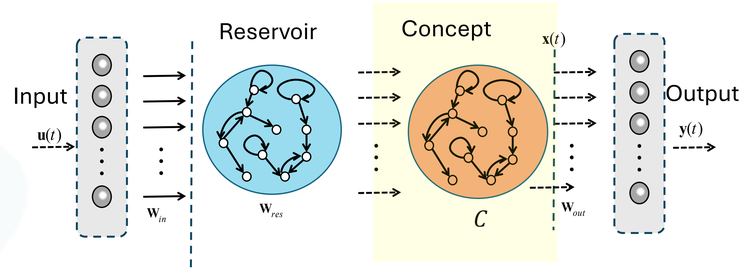
3. Our Approach

- We propose a Multiple-Model Mechanism based on Concept Echo State Networks (CESNs).
- Each CESN models a specific appliance's time-series consumption pattern within a building.
- The outputs of multiple CESNs are fused to predict the total building energy consumption.
- Each CESN encodes distinct usage behaviors, ensuring robustness and accuracy even with severe missing inputs.
- A Multi-Objective Optimization process strategically combines domain-specific CESNs to balance:
  - Prediction accuracy, and
  - Feature reconstruction quality.
- This fusion enhances global robustness and model interpretability, achieving reliable forecasts under data volatility.

4. Highlights

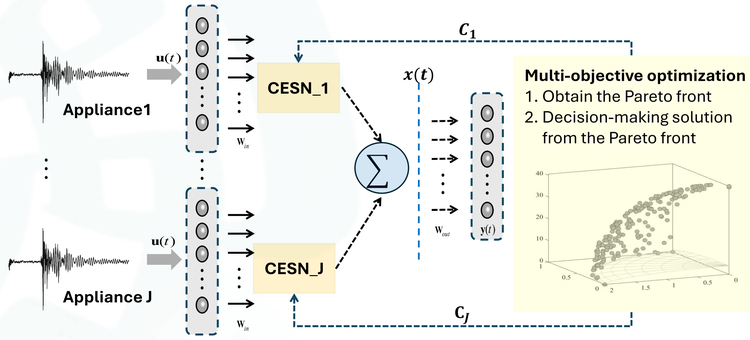
✅ Novel CESN Framework — modular, interpretable, and efficient for smart-building forecasting.
✅ Resilient to Missing Data — maintains high accuracy even under incomplete or corrupted inputs.
✅ Multi-Objective Optimization — simultaneously improves forecasting precision and feature recovery.
✅ Concept-Based Modeling — each model corresponds to an identifiable appliance or consumption "concept."
✅ Practical Impact — scalable to diverse Food-Energy-Water (FEW) systems, enabling more sustainable and explainable AI solutions.

Concept Echo State Network (CESN)



$$x(t+1) = C \cdot \varphi\left(W_{res}x(t) + W_{in}\,u(t)\right)$$

Multi-Concept ESN (MCESN)



$$min_{v\in\Omega}F(v,i) = \left(f^1(v,i), f^2(v,i), \ldots, f_m(v,i)\right)^T, s.t.\, v \in \Omega,$$

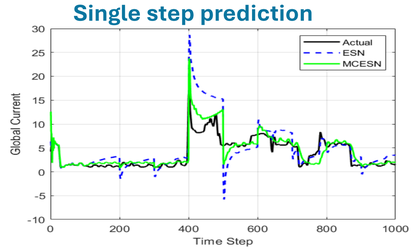
$$L^1(C) = \sum_{t=1}^{N} \|Cx(t) - x(t)\|^2 + \lambda C\|^2, L^2(C) = \sum_{t=1}^{N} \left\|Cx(t) - x_{target}(t)\right\|^2 + \lambda\|C\|^2$$

Experiments

We conducted experiments using the **UCI Household Electricity Consumption** dataset. This dataset records household electricity usage from **January to June 2007**, comprising approximately **260,640 entries**.
It includes measurements from **three sub-metered areas**—the **kitchen**, **laundry room**, and **water heater/air conditioner**—and exhibits an overall **25% missing data rate**, making it suitable for testing model robustness under incomplete conditions.

Github

Single step prediction



Performance comparison

Model	MAE	MSE	RMSE
LSTM	0.2762	0.5603	0.7485
GRU	0.3011	0.6842	0.8272
ESN	0.1996	0.3832	0.6190
DESN	0.1173	0.0837	0.2893
MCESN	**0.1311**	**0.0683**	**0.2614**

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